

# Jackett Auto-Configuration + Clean-Slate Rebuild “ Design

Revision: 1 Last modified: 2026-06-06T00:00:00Z

Date: 2026-04-26 Author: brainstorming session (Claude + project owner) Status: approved, ready for implementation plan Implementation skill next: superpowers:writing-plans

## 1. Problem & Intent

The project owner wants three things in one pass:

- 1. **Clean-slate rebuild** of the Python-profile container stack so the system boots from a known-zero state.
- 2. **Full Jackett indexer integration** “ every credentialed tracker we have an .env triple for should be auto-configured in Jackett at proxy startup, with no manual Jackett UI clicks. New trackers added to .env later should pick themselves up automatically.
- 3. **Comprehensive tests + challenges** at every CONSTITUTION-mandated layer, plus a parity audit so the eventual Go-only flip doesn’t silently drop features.

Closing instruction: commit and push to all upstreams as work progresses; hand off when the stack is verified and ready for manual testing.

## 2. Decisions Locked During Brainstorming

| #     | Question                               | Choice                                             | Implication                                                                                                                                                                                                                  |
|-------|----------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1    | What does “all Jackett trackers” mean? | (c) Auto-enable + credential injection from .env   | We write code that configures Jackett on the user’s behalf for indexers we have creds for. We do <b>not</b> auto-enable Jackett’s 600+ public indexers blindly.<br>RuTracker, Kinozal, NNMClub, IPTorrents today; any future |
| Q2    | Which credentialed indexers?           | (b) Generic env discovery + fuzzy match + override | <NAME>_USERNAME/_PASSWORD/_COOKIES triple gets picked up automatically; JACKETT_INDEXER_MAP resolves ambiguous matches.                                                                                                      |
| Q3    | What gets wiped on “clean slate”?      | (b) Containers + images + ./config/jackett         | Keep ./config/qBittorrent, ./tmp, /mnt/DATA. Forces auto-config to bootstrap from zero, which is the test we want.                                                                                                           |
| Q4/Q5 | Python or Go for the rebuild?          | (d) Python now, parity audit doc, Go port deferred | Implementation runs against Python proxy. A docs/migration/PARITY_GAPS.md deliverable catalogs what’s still unported                                                                                                         |

| #  | Question               | Choice                                                | Implication                                                                                                                                                                        |
|----|------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q6 | Test coverage floor?   | (a) Full CONSTITUTION compliance                      | in Go so the eventual flip loses nothing.<br>All 7 layers: unit, integration, E2E, security/penetration, benchmark, automation (mapped to tests/contract/ â€” see Â§5), challenge. |
| Q7 | When does push happen? | (c) Per-layer commit + push                           | After each test layer goes green, commit + push to all 3 remotes.                                                                                                                  |
|    | Approach Module shape? | (1) Module-in-merge_service, idempotent, startup-time | Lowest topology change, highest testability, easy to lift-and-shift to Go later.                                                                                                   |

## 3. Architecture

### 3.1 Module

**New file:** download-proxy/src/merge\_service/jackett\_autoconfig.py

**Public API (single function):**

```

async def autoconfigure_jackett(
    jackett_url: str,
    api_key: str,
    env: Mapping[str, str],
    indexer_map_override: dict[str, str] | None = None,
    timeout: float = 10.0,
) -> AutoconfigResult: ...

```

AutoconfigResult is a Pydantic model with: - ran\_at: datetime - discovered\_credentials: list[str] â€” tracker names found in env (NEVER includes the values) - matched\_indexers: dict[str, str] â€” env name â†’ Jackett indexer ID - configured\_now: list[str] â€” indexer IDs configured this run - already\_present: list[str] â€” indexer IDs Jackett already had configured (idempotency hits) - skipped\_no\_match: list[str] â€” env names where fuzzy match failed - skipped\_ambiguous: list[dict] â€” env name + candidate IDs where â‰¥2 indexers matched - errors: list[str] â€” structured error codes (e.g., jackett\_unreachable, jackett\_auth\_failed, catalog\_parse\_failed, indexer\_config\_failed:<id>)

The function never raises out of its boundary; all failures land in errors.

### 3.2 Call site

download-proxy/src/main.py â€” after JACKETT\_API\_KEY is verified non-empty (the existing startup already waits for Jackettâ€™s key extraction), and before the FastAPI app starts serving traffic. Failure is logged WARNING; **boot continues**. Auto-config is best-effort.

### 3.3 Read endpoint

GET /merge/jackett/autoconfig/last returns the cached AutoconfigResult from the most recent run. Read-only. Lives in download-proxy/src/api/routes.py. Response shape:

```
{
  "ran_at": "2026-04-26T14:23:11Z",
  "discovered": ["RUTRACKER", "KINOZAL", "NNMCLUB", "IPTORRENTS"],
  "configured_now": ["rutracker", "kinozalbiz"],
  "already_present": ["nnmclub"],
  "skipped_no_match": ["IPTORRENTS"],
  "skipped_ambiguous": [],
  "errors": []
}
```

Credentials NEVER appear in the response, the cached object's `__repr__`, or any log line at any level.

### 3.4 Discovery algorithm (3 passes)

1. **Env scan** regex `^([A-Z][A-Z0-9_]+?)_(USERNAME|PASSWORD|COOKIES)$` over `os.environ`. Group by tracker name. A "credential bundle" is valid when the bundle contains either (USERNAME && PASSWORD) or COOKIES. Names matching the **denylist** (default QBITORRENT, JACKETT, WEBUI, PROXY, MERGE, BRIDGE, overridable via `JACKETT_AUTOCONFIG_EXCLUDE` comma-separated env var) are dropped before pass 2 these are project-internal credentials (qBit auth, Jackett's own key, etc.) that are guaranteed not to be tracker creds.
2. **Indexer match** for each surviving bundle:
  - Lookup tracker name in `indexer_map_override` first (highest precedence).
  - If unmatched, fuzzy-match using `Levenshtein.ratio()` (threshold  $\approx 0.85$ , case-insensitive) against `Jackett/api/v2.0/indexers/all/results` catalog.
  - Ambiguous matches ( $\approx 2$  indexers  $\approx$  threshold) are skipped with a clear log + entry in `skipped_ambiguous`.
3. **Configure** for each matched indexer:
  - GET `/api/v2.0/indexers` (already-configured list). If indexer ID is present, add to `already_present`, skip.
  - GET `/api/v2.0/indexers/{id}/config` to fetch the field template Jackett expects.
  - Map our credential fields onto the template (e.g., `username`  $\mapsto$  `username`, `password`  $\mapsto$  `password`, `cookieheader`  $\mapsto$  `COOKIES` value).
  - POST `/api/v2.0/indexers/{id}/config` with the populated template. On 4xx, log + add to `errors`. On 5xx, one retry with 2s backoff via existing `tenacity`.

### 3.5 No new dependencies

Levenshtein, aiohttp, pydantic, tenacity are already in `download-proxy/requirements.txt`.

## 4. Data Flow

container boot

```
â"”â"€> start-proxy.sh waits for JACKETT_API_KEY in env
â"”â"€> python -m main
â"œâ"€> startup hook: autoconfigure_jackett(...)
â",      â"œâ"€ scan os.environ [pass 1]
â",      â"œâ"€ GET /api/v2.0/indexers/all/results (catalog)
â",      â"œâ"€ fuzzy-match bundles â†’ indexer IDs [pass 2]
â",      â"œâ"€ GET /api/v2.0/indexers (already-)
â",      â"œâ"€ GET /api/v2.0/indexers/{id}/config (template)
â",      â"œâ"€ POST /api/v2.0/indexers/{id}/config (apply, p
â",      â"”â"€ build AutoconfigResult, log summary,
â",      store in module-level cache
â"œâ"€> FastAPI app starts, exposes:
â",      GET /merge/jackett/autoconfig/last (cached, reda
â"”â"€> proxy + merge service serve traffic normally
```

**Persistence:** Jackett owns the durable state on disk (. /config/jackett/Jackett/Indexers/\*.json). Auto-config issues API calls; restart-survival is Jackettâ€™s job; the idempotency check on next boot prevents double-config.

**Result caching:** in-process, module-level. Lost on restart; re-run on next boot. No new files, no DB.

**Logging:** structured (existing project logger). One INFO summary per run (autoconfigured 3/4 discovered indexers). Per-indexer DEBUG. WARNING on errors. **No credential value at any level, ever.**

## 5. Test Matrix (CONSTITUTION 7-layer floor)

| # Layer                    | File                                                                          | Real infra?                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Unit                     | tests/unit/merge_service/<br>test_jackett_autoconfig.py                       | Mocks via<br>unittest.mock.AsyncMock/<br>MagicMock (matches existing convention)<br>in tests/unit/<br>test_private_tracker_search.<br>allowed in unit only) |
| 2 Integration              | tests/integration/<br>test_jackett_autoconfig_real.py                         | Real Jackett container via pytest-docker                                                                                                                    |
| 3 E2E                      | tests/e2e/<br>test_jackett_autoconfig_e2e.py                                  | Full stack (podman compose up)                                                                                                                              |
| 4 Security/<br>penetration | tests/security/<br>test_jackett_autoconfig_secrets.py<br>+ bandit static scan | Real stack + log files                                                                                                                                      |
| 5 Benchmark                | tests/benchmark/<br>test_jackett_autoconfig_perf.py                           | Real Jackett or recorded responses (allowed for benchmark stability)                                                                                        |
| 6 Automation               | tests/contract/<br>test_jackett_autoconfig_contract.py<br>(Schemathesis)      | Live FastAPI                                                                                                                                                |
| 7 Challenge                |                                                                               | Full live system                                                                                                                                            |

| # Layer | File                                                     | Real infra? |
|---------|----------------------------------------------------------|-------------|
|         | challenges/scripts/<br>jackett_autoconfig_clean_slate.sh |             |

## 5.1 Layer-by-layer assertions

**Unit (1):** - Env-scan regex correctness across edge cases (lowercase ignored, mixed case, multi-underscore tracker names). - Fuzzy threshold behavior “at, above, and below 0.85. - JACKETT\_INDEXER\_MAP override precedence over fuzzy match. - Idempotency check skips already-configured indexers. - `AutoconfigResult.__repr__()` and `.model_dump_json()` redact passwords/cookies. - Errors are caught at function boundary “autoconfigure\_jackett() never raises.

**Integration (2):** - Spin up a real Jackett container (`pytest-docker` fixture). - Inject test env vars (`TESTTRACKER_USERNAME=u, _PASSWORD=p`). - Call `autoconfigure_jackett()`, assert it discovered the env, fuzzy-matched a public indexer Jackett ships with, configured it. - Second call: `indexer_in_already_present`. - `/merge/jackett/autoconfig/last` returns the live result.

**E2E (3):** - Tear down stack, wipe `./config/jackett`, set fake creds in env, `podman` compose up, wait for healthchecks. - Poll `/merge/jackett/autoconfig/last` until populated. - Assert expected indexer set configured. - Run a search through the merge service. - Assert at least one Jackett-sourced result appears in the dedup output.

**Security (4):** - Boot stack with creds in env. Capture stdout/stderr from boot. - Force errors (invalid Jackett key, malformed env) and capture exception tracebacks. - Assert: no credential value appears in boot logs, FastAPI access logs, `/merge/jackett/autoconfig/last` response, exception tracebacks, or `./config/jackett/Jackett/log.txt`. - Run `bandit` against `merge_service/jackett_autoconfig.py`. Zero high-severity findings.

**Benchmark (5):** - `pytest-benchmark` baselines: - Env-scan over 1k env vars. - Fuzzy match against a 50-indexer catalog. - Full happy-path `autoconfigure` round-trip. - Baselines committed to `tests/benchmark/baselines/jackett_autoconfig.json`. - Fail if any benchmark regresses > 2%—baseline.

**Automation/Contract (6):** - `Schemathesis` loads the OpenAPI schema for `/merge/jackett/autoconfig/last`. - Fuzzes parameters; asserts no 5xx, no schema violations. - Documented mapping: `CONSTITUTION` says “automation” but `tests/` has no `automation/` directory. We map to `tests/contract/` (the closest existing equivalent “Schemathesis-driven contract testing of the public API surface). This mapping is recorded here so future readers don’t search for a missing dir.

**Challenge (7):** - `challenges/scripts/jackett_autoconfig_clean_slate.sh`:  
1. `podman compose down --remove-orphans`  
2. `rm -rf ./config/jackett`  
3. `podman compose up -d`  
4. Wait for healthchecks (timeout 3 min).  
5. Poll `/merge/jackett/autoconfig/last` until `ran_at` populated.  
6. Assert configured indexer count ≥ 1.  
7. Run a search via the merge service API.  
8. Assert results contain at least one Jackett-sourced entry.  
9. Exit 0 on success, non-zero with diagnostics on failure.

## 5.2 Resource caps

CLAUDE.md mandatory #9 caps tests at 30-40% host. All test invocations wrapped:

```
nice -n 19 ionice -c 3 -p 1 python3 -m pytest ...
```

Container `mem_limit` already enforces in-container caps. `pytest -p no:randomly --maxfail=1 -x` for deterministic per-layer runs.

## 6. Error Handling

| Failure                                                        | Behavior                                                                         | Visibility                                               |
|----------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------|
| Jackett unreachable at startup                                 | Log WARNING, <code>errors=["jackett_unreachable"]</code> , <b>boot continues</b> | <code>/merge/jackett/autocfg/last</code> shows the error |
| Jackett 401 (bad/missing key)                                  | Log WARNING <code>jackett_auth_failed</code> , no retries                        | Same                                                     |
| Catalog response malformed                                     | Log WARNING <code>catalog_parse_failed</code> , boot continues                   | Same                                                     |
| Credential triple incomplete (USERNAME only)                   | Skip silently at DEBUG â€” env may have unrelated vars                           | DEBUG only                                               |
| Fuzzy match ambiguous                                          | Skip with WARNING listing candidates, suggest <code>JACKETT_INDEXER_MAP</code>   | <code>skipped_ambiguous</code> entry                     |
| Fuzzy match below threshold                                    | Skip with INFO                                                                   | <code>skipped_no_match</code> entry                      |
| <code>JACKETT_INDEXER_MAP</code> references unknown indexer ID | Log WARNING, skip mapping                                                        | <code>errors</code> entry                                |
| Indexer config POST 4xx                                        | Log status + indexer ID (no body)                                                | <code>errors</code> entry                                |
| Indexer config POST 5xx                                        | One retry (2s backoff via <code>tenacity</code> ), then give up                  | <code>errors</code> entry                                |
| Indexer already configured                                     | Skip silently                                                                    | <code>already_present</code>                             |
| Network timeout mid-call                                       | Treated as 5xx for that one indexer; others continue                             | Per-indexer error                                        |

**Universal invariants:** - `autoconfigure_jackett()` never raises out of its boundary. - Total runtime hard-capped at 60s via outer `asyncio.wait_for`. - Credentials never appear in logs, error messages, the read endpoint, or exception text.

## 7. Parity Audit Deliverable

**File:** `docs/migration/PARITY_GAPS.md`

**Method:** side-by-side read of `download-proxy/src/{api,merge_service,ui,config}` vs `qBittorrent-go/internal/{api,service,client,models,middleware,config}`. One row per Python public function / endpoint / behavior. No code changes. Estimated 1-2 hours.

**Document shape (excerpt):**

```
# Python â†’ Go Parity Gaps
```

```
Last audited: 2026-04-26 (commit <sha>)
```

Audit method: side-by-side read of public API surface; no behavior testing

### ## Summary

- N Python features audited
- X fully ported
- Y partial
- Z missing

### ## Matrix

| Python module                | Feature               | Go location  | Status                       | Risk if Go-only today |
|------------------------------|-----------------------|--------------|------------------------------|-----------------------|
| `merge_service/search.py`    | `start_search()`      | orchestrator | `internal/service/search.go` | Missing               |
| `merge_service/enricher.py`  | TMDB title resolution | (none)       | Missing                      | Missing               |
| `merge_service/validator.py` | BEP 48 HTTP scrape    | (none)       | Missing                      | Missing               |
| ...                          | ...                   | ...          | ...                          | ...                   |

### ## Per-gap follow-up specs (proposed, in priority order)

1. Port `enricher.py` to `internal/service/enricher.go`
2. Port `validator.py` (BEP 48 + BEP 15)
3. ...

**Status definitions** (locked): - **Ported** “feature exists in Go with equivalent behavior. - **Partial** “feature exists in Go but lacks a sub-behavior. - **Missing** “no Go counterpart.

## 8. Verification & Push Protocol

**Order of operations** (each step is a gate; failure halts):

| # | Step                                                            | Commit + push? |
|---|-----------------------------------------------------------------|----------------|
| 1 | Pre-flight: clean tree, env vars present                        | “              |
|   | Tear down: containers + images + ./config/jackett.              |                |
| 2 | Keep ./config/qBittorrent, ./tmp, /mnt/DATA                     | “              |
|   | Implement auto-config module + wire into main.py + add endpoint |                |
| 3 |                                                                 | “              |
| 4 | Layer 1 (Unit) green                                            | yes            |
|   | Boot real stack (podman compose up -d), wait for healthchecks   |                |
| 5 |                                                                 | “              |
| 6 | Layer 2 (Integration) green                                     | yes            |
| 7 | Layer 3 (E2E) green                                             | yes            |
| 8 | Layer 4 (Security) green                                        | yes            |

| #  | Step                                                                                                                                     | Commit + push? |
|----|------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 9  | Layer 5 (Benchmark)<br>green, baselines<br>committed                                                                                     | yes            |
| 10 | Layer 6 (Automation/<br>Contract) green                                                                                                  | yes            |
| 11 | Layer 7 (Challenge)<br>script exits 0                                                                                                    | yes            |
| 12 | Parity audit: read both<br>codebases, write<br>PARITY_GAPS.md                                                                            | yes            |
| 13 | Final clean-slate<br>verification: full tear-<br>down + boot from<br>zero, run challenge<br>again, capture output<br>into commit message | yes            |
| 14 | Handoff message to<br>project owner                                                                                                      | â€”            |

**Push target:** `git push origin <branch> && git push github <branch> && git push upstream <branch>` after each gate. (All three URLs identical â€” push is idempotent on the remote, but we do all three so refs match locally.)

**Hard stops** (require owner input): - Any layer fails after one fix attempt. - Any push rejected. - Healthcheck unhealthy after 3 minutes. - Bench regression > 2%—baseline.

## 9. File Inventory

### 9.1 New files

- `download-proxy/src/merge_service/jackett_autoconfig.py` â€” the module
- `tests/unit/merge_service/test_jackett_autoconfig.py`
- `tests/integration/test_jackett_autoconfig_real.py`
- `tests/e2e/test_jackett_autoconfig_e2e.py`
- `tests/security/test_jackett_autoconfig_secrets.py`
- `tests/benchmark/test_jackett_autoconfig_perf.py`
- `tests/benchmark/baselines/jackett_autoconfig.json`
- `tests/contract/test_jackett_autoconfig_contract.py`
- `challenges/scripts/jackett_autoconfig_clean_slate.sh` (executable)
- `challenges/scripts/run_all_challenges.sh` (currently no challenges/ dir)
- `docs/migration/PARITY_GAPS.md`
- `docs/superpowers/specs/2026-04-26-jackett-autoconfig-clean-rebuild-design.md` (this file)

### 9.2 Modified files

- `download-proxy/src/main.py` â€” call `autoconfigure_jackett()` after Jackett key extraction, before `uvicorn`



- `download-proxy/src/api/routes.py` â€” add `GET /merge/jackett/`  
`autoconfig/last`
- `download-proxy/src/api/__init__.py` â€” register the new route if needed
- `docs/JACKETT_INTEGRATION.md` â€” append â€œAuto-Configurationâ€” section
- `CLAUDE.md` â€” add `JACKETT_INDEXER_MAP` and  
`JACKETT_AUTOCONFIG_EXCLUDE` to the env vars line; note the auto-config endpoint
- `AGENTS.md` â€” same env additions
- `.env.example` (if exists) â€” add `JACKETT_INDEXER_MAP=` and  
`JACKETT_AUTOCONFIG_EXCLUDE=` placeholders

### 9.3 Untouched

- `qBitTorrent-go/**` (parity audit reads but does not modify)
- `frontend/**`
- `plugins/**`
- `webui-bridge.py`
- `start-proxy.sh` (unless implementation discovers a timing tweak is needed)

### 9.4 Bugfix log

Per `CLAUDE.md` mandatory #10 + CONST-032: any bug discovered during this work goes into `docs/issues/fixed/BUGFIXES.md` with a corresponding `challenges/scripts/jackett_autoconfig_<bug-slug>.sh` regression guard.

## 10. Out of Scope (explicit)

- Porting any Python feature to Go. The parity doc *catalogs* the gap; closing it is future work in dedicated specs.
- Go-profile rebuild or Go-side Jackett auto-config.
- Wiping `./config/qBittorrent/` or `/mnt/DATA`.
- Manual testing scripts â€” handoff is verbal.
- Auto-enabling Jackettâ€™s public indexer set without credentials. The Q1 decision was credentialed-only.

## 11. Open Questions for Implementation Plan

(These do not block design approval; they get resolved when `writing-plans` produces the step-by-step plan.)

- Exact branch name for this work (suggest `jackett-autoconfig-clean-rebuild`).
- Whether the existing test stack already auto-starts Jackett via `pytest-docker`, or whether the integration layer needs a new fixture.
- Whether `docs/issues/fixed/BUGFIXES.md` exists â€” if not, it gets created on first bugfix.
- Whether `start-proxy.sh` injects `JACKETT_API_KEY` to the proxy container before `python -m main` runs (it should, but verify at implementation time).